

Project Development Phase

Preprocessing steps and Business Questions with visualizations Document

Date	06 July 2026
Team ID	
Project Name	Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau
Maximum Marks	

Data Preprocessing Steps

1. Data Cleaning

Handling Missing Values

- Verified that the dataset contained no missing (null) values.
- No records required removal or replacement.

Removing Duplicates

- Checked for duplicate records.
- No duplicate entries were found.

Data Consistency

- Verified that the values in all fields were consistent.
- No text standardization or formatting changes were required.

2. Data Transformation

Data Types

- Verified that all fields had the correct data types.
- No datatype conversions were required.

Calculated Field Creation

- Created a calculated field named **“All Skin Types”** to determine the total products suitable across all skin types.
- The calculated field was created using the formula:
[Combination] + [Dry] + [Normal] + [Oily] + [Sensitive]

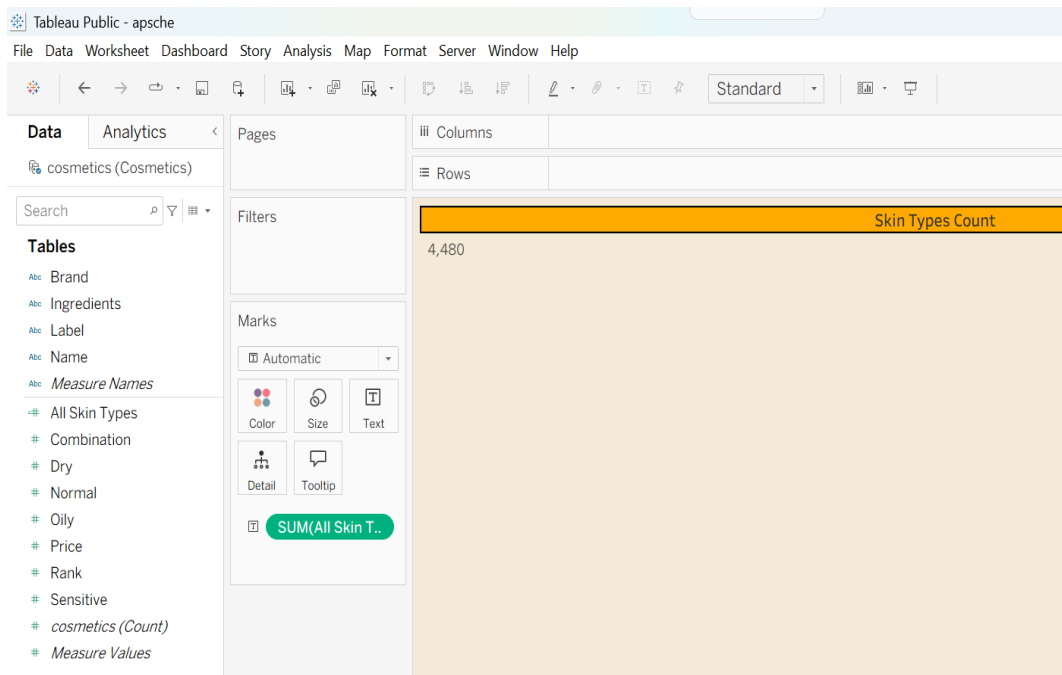
Data Preparation

- The dataset was already clean and structured.

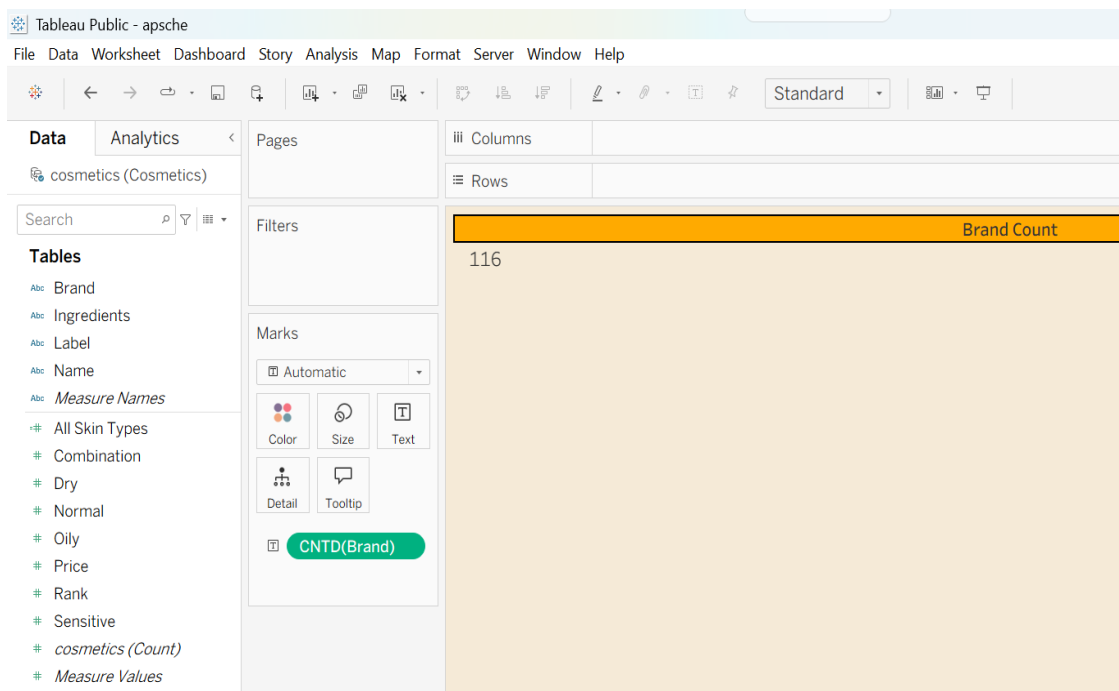
- No merge, split, or additional transformation operations were required before importing it into Tableau.

Business Questions with visualizations

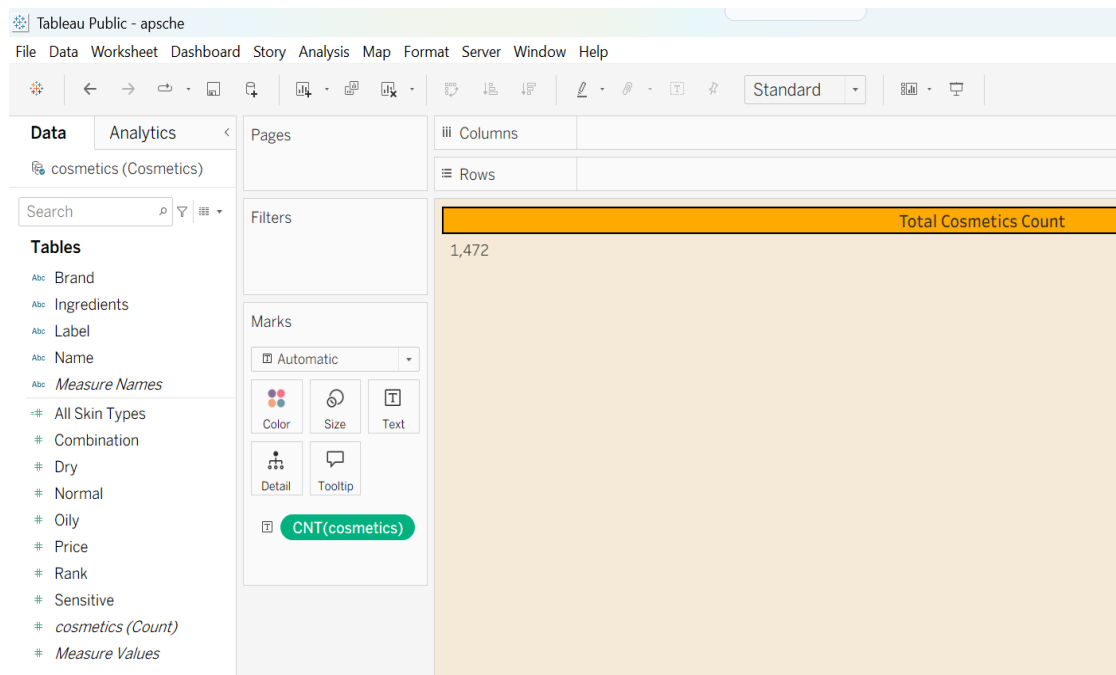
1. What is the total skin-type count across all cosmetic products?



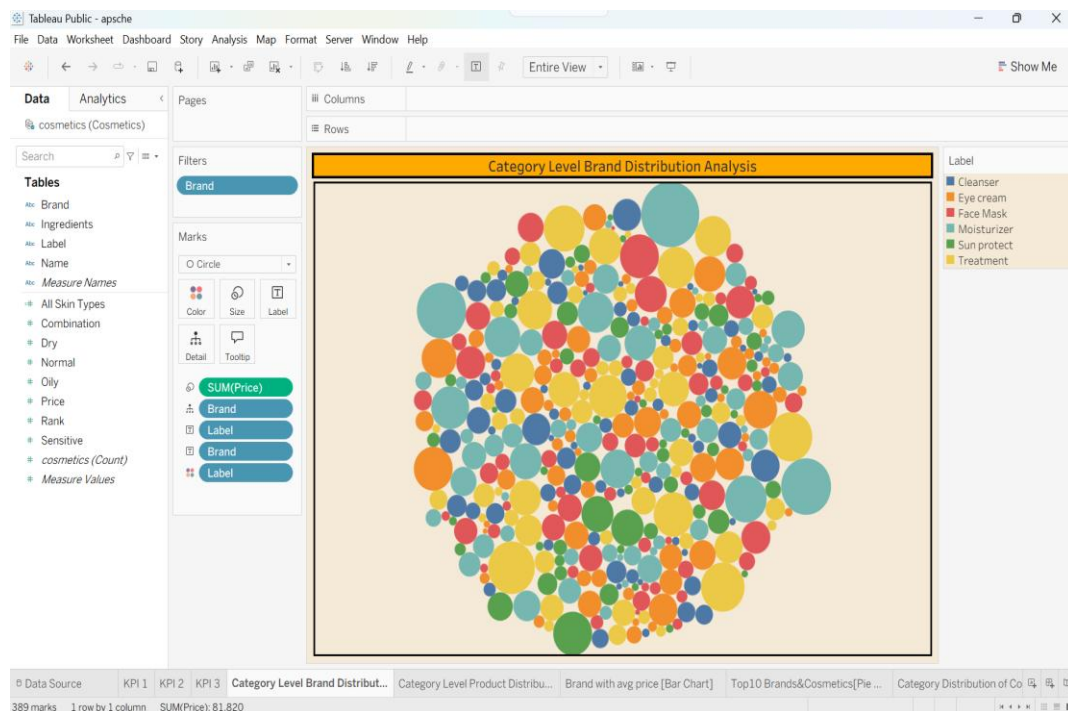
2. How many distinct cosmetic brands are available in the dataset?



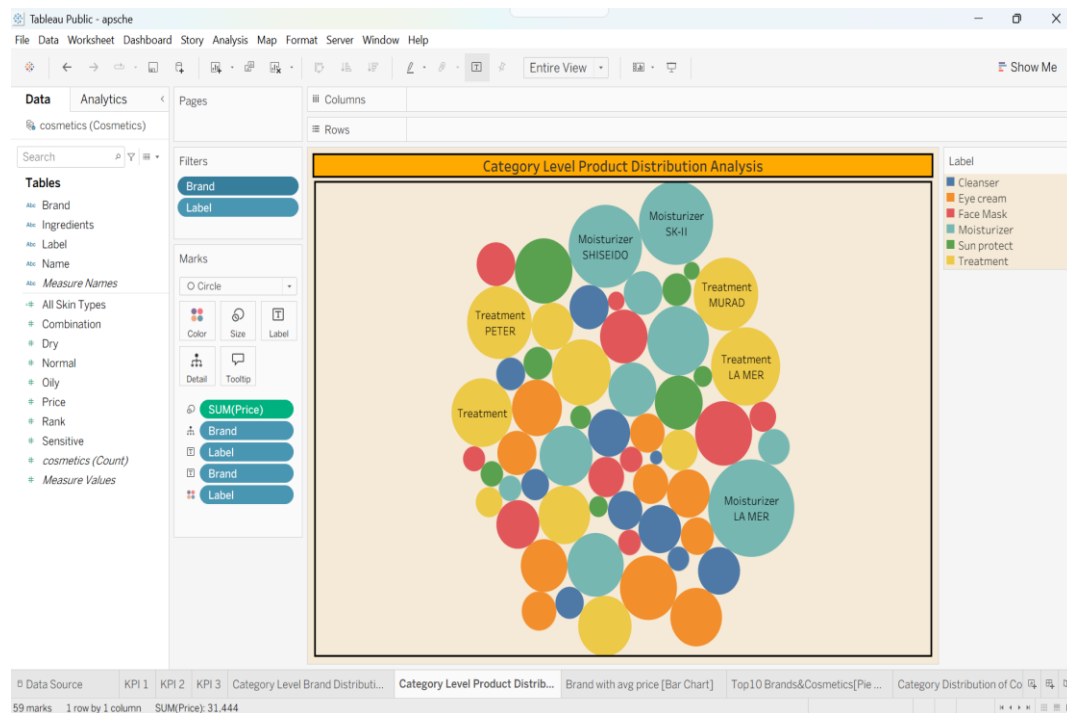
3. What is the total number of cosmetic products available in the dataset?



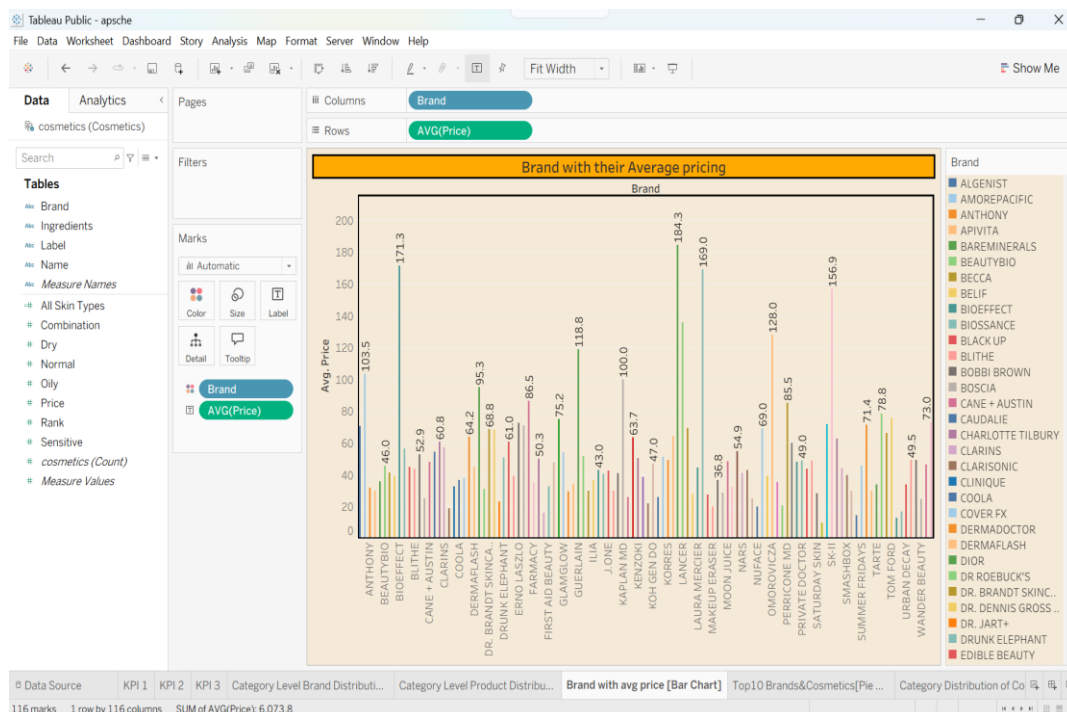
4. How are cosmetic brands distributed across different product categories?



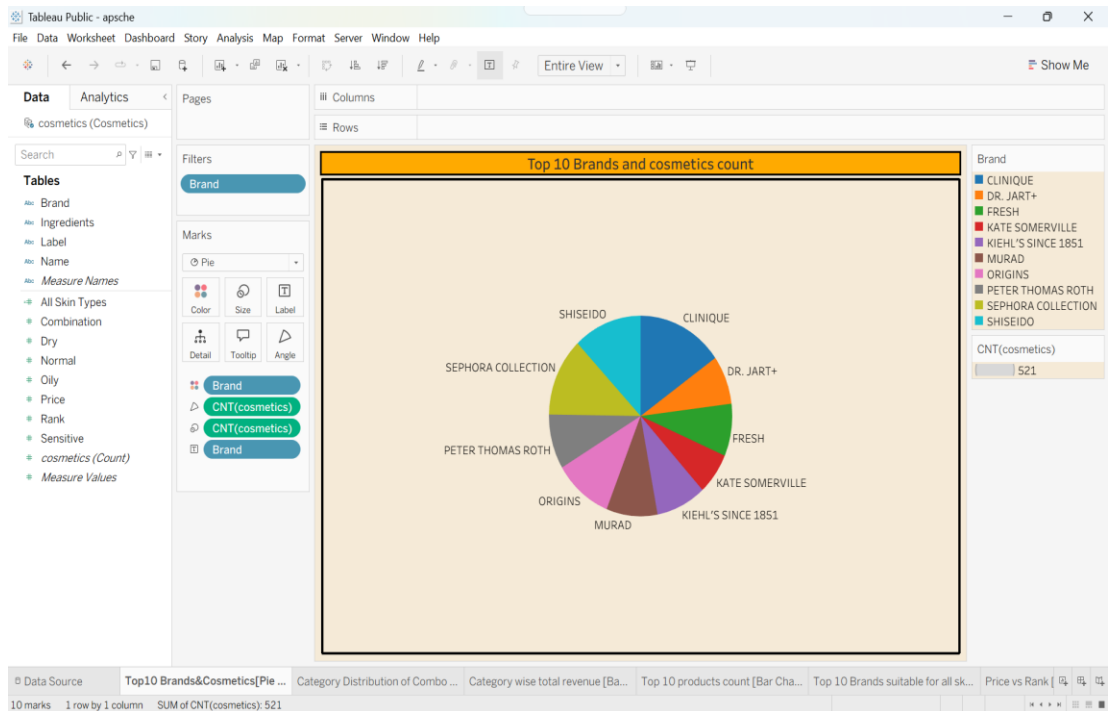
5. How are cosmetic products distributed across different product categories for the top 10 brands?



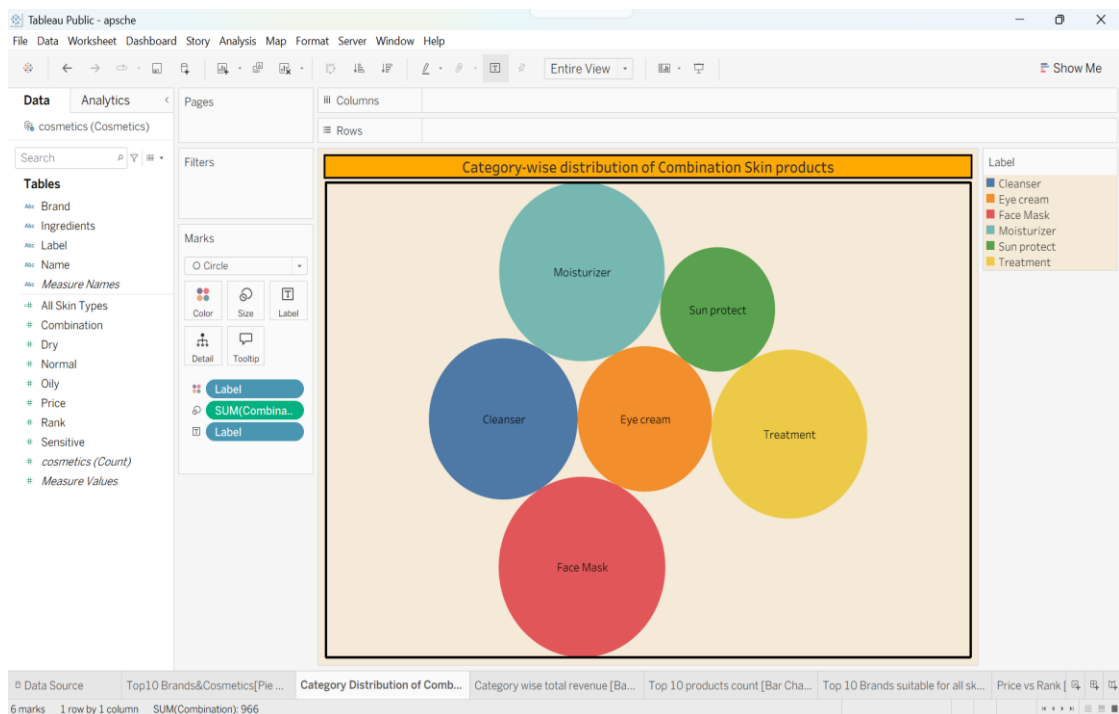
6. How does the average price vary across different cosmetic brands?



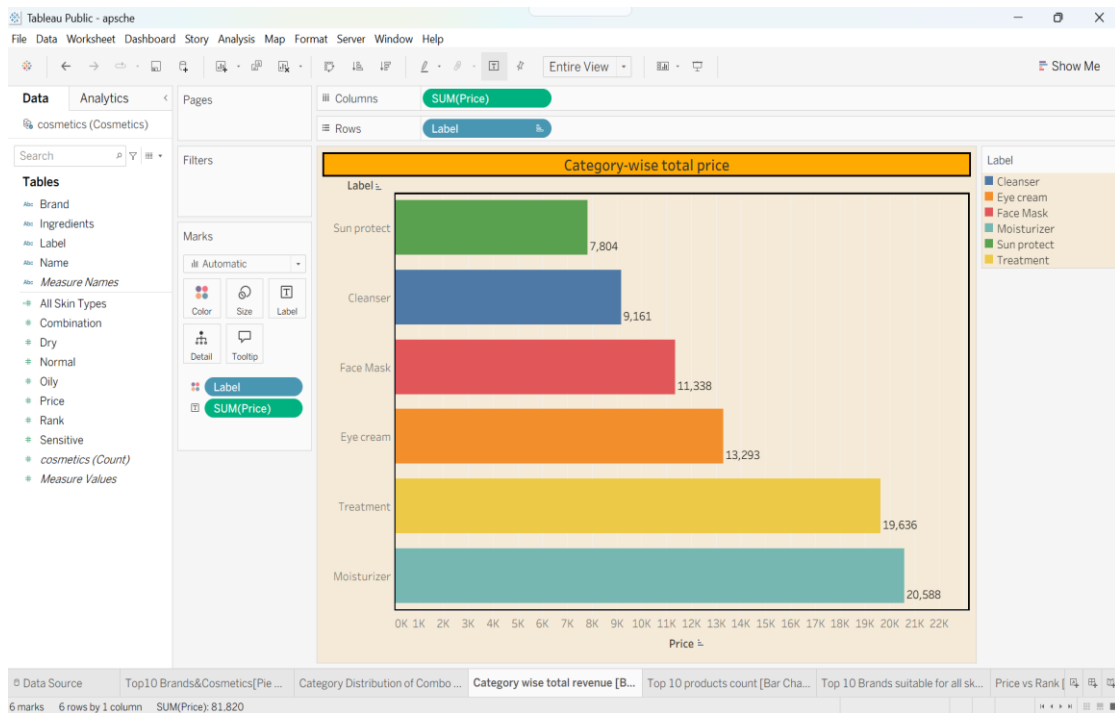
7. How are cosmetic products distributed among the top 10 cosmetic brands?



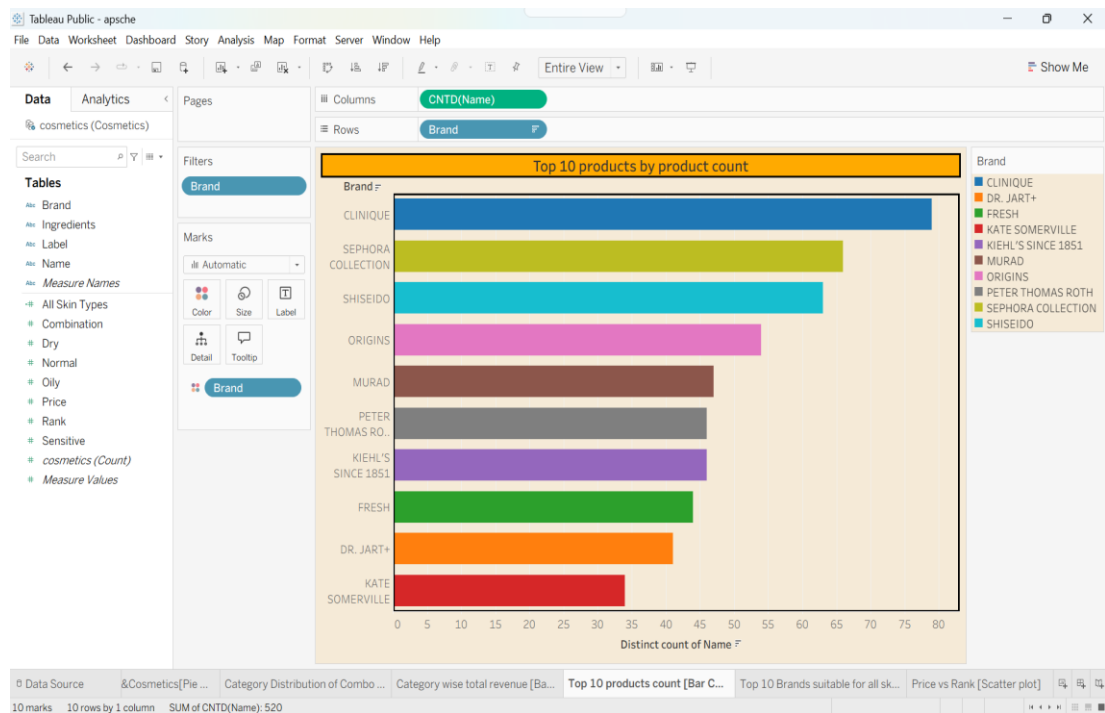
8. How are products suitable for combination skin distributed across different cosmetic categories?



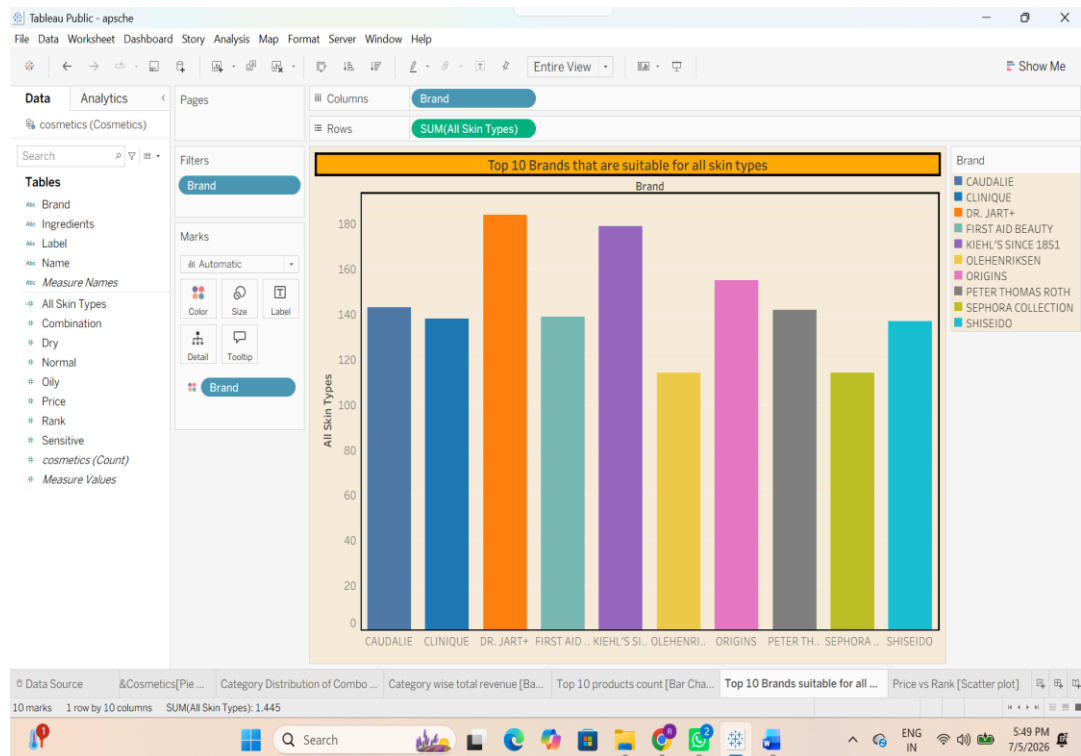
9. Which cosmetic category has the highest total price of products?



10. Which top 10 cosmetic brands offer the highest number of distinct products?



11. Which top 10 cosmetic brands offer the highest number of products suitable for all skin types?



12. Is there a relationship between product price and product rank across cosmetic brands?

